

On the opportunities and risks of foundation models

(Bommasani et al., 2022)

foundation model = trained on broad data and that can be repurposed for many different use cases

emergence = the ability of models to perform tasks that they were not specifically trained for

homogenization = the same tool/model being used for many different purposes, meaning that those purposes are subject to any problems in the foundation model

self-supervised learning trains without the need for annotated data (and annotating is expensive and time-consuming, apparently)

almost all major NLP models (basically GenAI with words?) have been adopted from earlier foundation models

emergence can be a problem because if the tool wasn't trained for the purpose, it's hard to troubleshoot if/when things go wrong

understanding foundation models requires looking at the different stages in the ecosystem model of AI

academia can and could play an important role in the development of AI since they're existing institutions that house a wide range of experts in many different disciplines

existing economic incentives mean that the big AI firms don't really address concerns/problems that won't make them (a lot of) money

they're also likely to ignore externalities since they don't pay the cost for them

and they have no incentive to have an open, transparent system

at one point, there was a lot of openness in model training, but that's changed and makes it a lot harder to conduct academic/independent research

“the suitability of a technology relying largely on emergent behavior for widespread deployment to people is unclear.” (p. 12)

AI Index report

Executive Summary

(lots of headline AI updates)

Chapter 4: Economy

highlights:

- global corporate AI investment more than doubled
- the US maintains a huge lead in global private AI investment
- AI companies are earning record revenues, but costs are also at record levels
- estimated consumer surplus from GenAI is up 54% annually
- most organizations now use AI in one way or another
- GenAI adoption has been very fast by historical standards
- AI is impacting the labor market, although unevenly
- a third of organizations think AI will reduce their workforce this year
- AI productivity gains are biggest in structured, measurable work where you can easily monitor outputs (shocking)
- China is still way ahead in industrial robots

investment in AI is heavily concentrated in a few companies in a few countries

53% “population-level adoption,” whatever that means

impact on labor is very context-dependent - really depends on the work you’re trying to do

AI may actually help more junior workers by helping to close skill gaps (but how does that square with them being hired less?)

major workforce effects have yet to show up, although some small reductions in force are expected over the next year

workers would be okay with AI automating away some of the more repetitive tasks they face if that frees them up to do more meaningful work

collaborative robots are gaining market share over traditional robots

Chapter 7: Education

AI learning and teaching is lagging behind innovation

highlights:

- fewer students in computer science but more in AI
- the US still leads in graduates in information, communications, and technology graduates
- ~80% of HS and college students use AI for schoolwork, but school policies aren't keeping up
 - about half of middle and high schools have policies
 - only 6% of teachers say their school's AI policies are clear
 - most common tasks are research, editing writing, and brainstorming
- mandatory AI education isn't common in schools, though some countries (notably China and the UAE) have mandated it nationally
- AI PhDs in the US have increased, but the increased number are going to academia instead of industry, which is a change
- AI skills are being acquired outside of formal education and some see them worth including on their resumes

generally, it's difficult to assess the state of AI in education because of a lack of data

there's wide agreement that AI literacy needs to be taught, but discourse often doesn't distinguish AI in education from AI education and AI literacy

AI literacy = foundational understanding of AI: how it works, how to use it, risks of usage

AI education = technical skills to build AI systems (not the definition I'd give)

AI in education = using AI tools in teaching and learning

Postsecondary CS & AI Education

fewer students in computer science but more in AI

67% of US university students report using generative AI for schoolwork, up from 20% in 2023

most highly cited reasons for non-use (>40% of respondents):

- concerns about academic misconduct
- concerns about content accuracy
- school policies constraining AI use

both high school and college students are most likely to use AI for research, brainstorming, and editing writing, although college students are more likely to use it to help them understand concepts

students seem to use AI more for higher-order than lower-order skills, which may indicate they're using it as a crutch

generally, students think AI has positively impacted their learning experience

K-12 CS and AI Education

only four states include AI in their computer science standards (who has CS standards?!)

60% of US schools offer computer science (!), but unsurprisingly it's more common in larger schools

K-12 students seem somewhat supportive in general of AI use for schoolwork

only about half of schools have AI policies, and they vary (e.g., can you use it at all)

30 states have created guidance on AI in education

state CS standards vary in the extent to which they include AI

new K12 CS standards (guidelines/suggestions?) will be forthcoming in summer 2026

most state AI guidelines are non-binding and implemented at local levels